51.2 V · 100 AH · 5.12 KWH · LITHIUM IRON PHOSPHATE

OLB-5.12 Series Storage Battery

A safe, long-life LFP battery pack for homes, small offices and micro off-grid systems. Available wall-mount or 19-inch rack-mount, and expandable to 10 packs in parallel for up to 51.2 kWh — a natural partner to the Olitec OLI-1H hybrid inverter.

OLB-5.12W · Wall-mount | OLB-5.12R · Rack-mount (3U)

Key figures

Rated energy	**5.12 kWh per pack**
Nominal voltage	**51.2 V**
Capacity	**100 Ah**
Max. charge / discharge	**100 A continuous**
Cycle life	**4,000 @ 80 % DoD**
Parallel packs	**up to 10 (51.2 kWh)**
Communication	**RS485, CAN, RS232**

- **$LiFePO_4$ safety** — thermally stable, cobalt-free cells certified to BIS, UN 38.3, IEC 62619 and UL 1973
- **Intelligent BMS** — cell-level voltage, current and temperature monitoring with full protection
- **Scales with demand** — start with one pack and parallel up to 10 on the same 51.2 V bus
- **5 kW continuous** — 50 A recommended, 100 A maximum per pack; 300 A peak for 10 s
- **Open communication** — RS485 × 3, CAN × 1, RS232 × 1 for inverter integration and monitoring
- **Flexible installation** — slim 198 mm wall-mount or standard 3U / 19-inch rack-mount; passive cooling

Parallel system specifications

Parameter	Specification
Nominal capacity	100 Ah × N (N = number of packs, max. 10)
Usable energy range	5.12 kWh to 51.2 kWh
Nominal voltage	51.2 V
Recommended charge / discharge current	50 A continuous
Max. charge / discharge current	100 A
Operating voltage	43.2 – 57.6 V
Operating temperature	0 – 50 °C (charging), 0 – 55 °C (discharging)
System weight	Number of packs × unit weight
Dimensions	As per general arrangement drawing / mounting layout
Parallel connection	Module power combiner box recommended for parallel systems

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Pack specifications

51.2 V LFP

Parameter	OLB-5.12W (Wall-mount)	OLB-5.12R (Rack-mount)
Nominal voltage	51.2 V	51.2 V
Nominal capacity	100 Ah	100 Ah
Rated energy	5.12 kWh	5.12 kWh
Operating voltage range	44.8 – 57.6 V	44.8 – 57.6 V
Recommended charge / discharge current	50 A continuous	50 A continuous
Max. charge / discharge current	100 A	100 A
Charging temperature range	0 °C to 50 °C	0 °C to 50 °C
Discharging temperature range	0 °C to 55 °C	0 °C to 55 °C
Cooling	Passive	Passive
Internal resistance	≤ 0.25 mΩ	≤ 0.25 mΩ
Cycle life	4,000 cycles @ 25 °C, 80 % DoD, 0.5C / 0.5C, to 70 % SoH	4,000 cycles @ 25 °C, 80 % DoD, 0.5C / 0.5C, to 70 % SoH
Dimensions (H × W × D)	644 × 410 × 198 mm	133.5 × 482 × 553 mm (3U, 19″ rack)
Weight	54 ± 2 kg	45 ± 2 kg
Ingress protection	IP20	IP20
Communication	RS485 × 3, CAN × 1, RS232 × 1	RS485 × 3, CAN × 1, RS232 × 1
Storage condition	15 – 35 °C; self-discharge ≈ 7 % / month; recharge every 3 months	15 – 35 °C; self-discharge ≈ 7 % / month; recharge every 3 months
Mounting	Wall-mounted	Rack-mounted

Cell specifications

Parameter	Specification
Cell chemistry	Lithium iron phosphate (LFP / $LiFePO_4$)
Nominal rating	3.2 V, 100 Ah (320 Wh per cell)
AC internal impedance	≤ 0.6 mΩ (0.3 mΩ typical)
Standard continuous charge / discharge current	50 A / 50 A (0.5C / 0.5C)
Max. continuous charge / discharge current	100 A / 100 A (1C / 1C)
Peak charge / discharge current (10 s, non-cyclic)	200 A / 300 A (2C / 3C)
Cell-level certification	BIS, UN 38.3, IEC 62619, UL 1973

Installation notes

- **Environment** — indoor installation (IP20); 0 – 50 °C charging, 0 – 55 °C discharging.
- **Paralleling** — use the module power combiner box when connecting two or more packs.
- **Clearance** — keep 100 mm around the pack for ventilation; passive cooling only.
- **Storage** — 15 – 35 °C; self-discharge about 7 % per month, recharge at least every 3 months.

Cycle life tested at 25 °C, 80 % depth of discharge, 0.5C / 0.5C to 70 % state of health. Specifications subject to change without notice.

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